

Intelligent Reflecting Surface Assisted Anti-Jamming Communications in THz Wideband Systems: A Fuzzy WoLF-PHC Learning Approach with Hybrid Beamforming

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Abstract—Wideband terahertz (THz) communications offer multi-gigahertz spectrum access but are acutely vulnerable to targeted jamming owing to their narrow, highly directional beams. This paper proposes a unified framework for intelligent reflecting surface (IRS)-assisted anti-jamming in THz wideband systems, jointly optimizing transmit power allocation, IRS phase shifts, and hybrid analog-digital beamforming against a smart adaptive jammer. The resulting optimization problem is non-convex, high-dimensional, and subject to a dynamic adversary with unknown strategy; we decompose it via a three-block alternating optimization (AO) framework in which RL handles discrete power control and low-complexity surrogate updates handle IRS configuration and beamforming. The novel fuzzy Win-or-Learn-Fast policy hill-climbing (WoLF-PHC) RL agent manages discrete power allocation using a 27-component fuzzy state abstraction and win/loss-adaptive learning rates, converging to a mixed strategy that is inherently unpredictable to a jammer monitoring action history. The THz system model incorporates $f_c = 100$ GHz, $B = 10$ GHz wideband OFDM (64 subcarriers), a 64-antenna sub-connected base station with $N_{\text{RF}} = 8$ RF chains and TTD-assisted beam-squint compensation, and a 256-element sub-array phase-dependent partial-connection (SPDP) IRS with per-sub-array true-time-delay units for wideband IRS beam-squint compensation. The IRS improves anti-jamming resilience by enhancing the desired effective channel and increasing the SINR margin under direct jammer interference. Evaluated over five independent seeds with 600 training episodes each, the proposed method achieves 11.87 bits/s/Hz system rate and 81.6% SINR protection, outperforming classical Q-learning (8.25 bits/s/Hz, 49.0%), DQN (8.23 bits/s/Hz, 49.1%), and the deterministic AO baseline (7.89 bits/s/Hz, 43.2%) with substantially lower seed variance. Comprehensive parameter sweeps over transmit power, IRS size, SINR threshold, and jammer power confirm consistent gains and a pronounced regime transition: deterministic optimization leads at low jammer power, whereas WoLF-PHC dominates under stronger adversarial pressure.

Index Terms—Anti-jamming, intelligent reflecting surface, terahertz communications, hybrid beamforming, reinforcement learning, WoLF-PHC, beam squint.

I. INTRODUCTION

THz systems provide massive bandwidth but are highly vulnerable to adaptive jamming because links are narrow-beam, frequency selective, and sensitive to beam squint. IRS-assisted anti-jamming is promising [6]–[8], but prior RL-based studies are largely narrowband and do not jointly address

wideband THz beam squint, sub-connected hybrid beamforming constraints, and smart jammers that exploit predictable control policies [1], [9], [10]. This work closes that gap with a unified THz-OFDM formulation and a robust mixed-strategy controller.

A. Contributions

The main contributions are:

- 1) **Joint THz anti-jamming formulation:** We jointly optimize power allocation, SPDP-IRS phase control, and sub-connected hybrid beamforming for a 100 GHz, 10 GHz-bandwidth THz-OFDM system with adaptive jamming. Unlike prior narrowband IRS anti-jamming studies, our formulation explicitly incorporates frequency-dependent THz channels, SPDP hardware coupling constraints, and TTD-assisted beam-squint compensation at both the IRS and BS.
- 2) **Three-block RL+AO solver:** We combine RL for discrete power control (42 actions, 7 fractions $\times$ 6 allocation modes) with low-complexity surrogate updates for SPDP-IRS phases (geometric AoA/AoD design) and SVD/MVDR updates for TTD-assisted hybrid beamforming.
- 3) **Fuzzy WoLF-PHC with real-data validation:** A 27-component fuzzy state abstraction and win/loss-adaptive policy updates improve robustness against policy-exploiting jammers; five-seed, 600-episode experiments and real sweeps show 44% rate and 66% protection gains over the best baseline, with the IRS improving anti-jamming resilience by strengthening the desired effective channel against direct jammer interference.

Notation: Bold lower/uppercase denotes vectors/matrices; $(\cdot)^H$, $|\cdot|$, $\|\cdot\|$, $\text{diag}(\cdot)$, and $\mathbf{1}\{\cdot\}$ have standard meanings.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. System Description

We consider a downlink THz-OFDM system comprising a $N = 64$ -antenna base station (BS) with $N_{\text{RF}} = 8$ RF chains, a 256-element (16×16) SPDP-IRS, $K = 4$ single-antenna users, and a colluding 2-antenna smart jammer, operating at $f_c = 100$ GHz with bandwidth $B = 10$ GHz and $M_{\text{sc}} = 64$

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OFDM subcarriers. The BS employs a sub-connected hybrid analog-digital architecture with TTD units in which each RF chain drives $N_{\text{per}} = N/N_{\text{RF}} = 8$ dedicated antennas, balancing hardware complexity with spatial degrees of freedom. The IRS is deployed to establish a reflected path between the BS and users, augmenting the direct link. Under the present model, the jammer interferes directly with each user and the IRS improves anti-jamming resilience by strengthening the desired effective channel, thereby increasing the SINR margin rather than directly suppressing the jammer signal.

B. Signal Model

Let $\mathbf{G}[m] \in \mathbb{C}^{M \times N}$ denote the BS-IRS channel at subcarrier m , $\mathbf{g}_{ru,k}[m] \in \mathbb{C}^M$ the IRS-to-user- k channel, $\mathbf{g}_{bu,k}[m] \in \mathbb{C}^N$ the direct BS-to-user- k channel, and $\mathbf{h}_{J,k} \in \mathbb{C}^{N_J}$ the jammer-to-user- k channel. The IRS introduces a diagonal reflection matrix $\Phi[m] \in \mathbb{C}^{M \times M}$ at subcarrier m ; the exact SPDP-constrained form of $\Phi[m]$ is given in (10) (Section III), and $\Phi[m]$ denotes this same matrix throughout.

The received signal at user k on subcarrier m is

$$y_k[m] = \underbrace{(\mathbf{g}_{ru,k}^H[m]\Phi[m]\mathbf{G}[m] + \mathbf{g}_{bu,k}^H[m]) \mathbf{w}_k[m] \sqrt{P_k} s_k}_{\text{desired signal}} + \underbrace{\sum_{i \neq k} (\mathbf{g}_{ru,k}^H[m]\Phi[m]\mathbf{G}[m] + \mathbf{g}_{bu,k}^H[m]) \mathbf{w}_i[m] \sqrt{P_i} s_i}_{\text{inter-user interference}} + \underbrace{\sqrt{P_{J,k}} \mathbf{h}_{J,k}^H \mathbf{z}_k j_k}_{\text{jamming}} + n_k \quad (1)$$

where $\mathbf{w}_k[m] \in \mathbb{C}^N$ is the beamforming vector, P_k is the allocated power, s_k is the unit-power data symbol, $P_{J,k}$ is the instantaneous jamming power toward user k , $\mathbf{z}_k$ is the jammer's spatial precoder, j_k is the jamming symbol, and $n_k \sim \mathcal{CN}(0, \sigma^2)$ is additive white Gaussian noise with per-subcarrier variance

$$\sigma^2 = k_B T \frac{B}{M_{\text{sc}}} \cdot 10^{F_{\text{NF}}/10} \quad (2)$$

where k_B is Boltzmann's constant, $T = 290$ K is the reference temperature, and $F_{\text{NF}} = 10$ dB is the receiver noise figure (converted to linear scale before multiplication).

C. SINR and System Rate

We define the effective channel row vector for user k at subcarrier m as

$$\mathbf{h}_{\text{eff},k}^H[m] \triangleq \mathbf{g}_{ru,k}^H[m]\Phi[m]\mathbf{G}[m] + \mathbf{g}_{bu,k}^H[m] \in \mathbb{C}^{1 \times N}. \quad (3)$$

This matches the receive-side combination in (1): the desired signal term is $\mathbf{h}_{\text{eff},k}^H[m] \mathbf{w}_k[m] \sqrt{P_k} s_k$.

The received SINR at user k on subcarrier m is

$$\text{SINR}_k[m] = \frac{P_k |\mathbf{h}_{\text{eff},k}^H[m] \mathbf{w}_k[m]|^2}{\sum_{i \neq k} P_i |\mathbf{h}_{\text{eff},k}^H[m] \mathbf{w}_i[m]|^2 + P_{J,k} |\mathbf{h}_{J,k}^H \mathbf{z}_k|^2 + \sigma^2}. \quad (4)$$

The per-user rate averaged over subcarriers is

$$R_k = \frac{1}{M_{\text{sc}}} \sum_{m=1}^{M_{\text{sc}}} \log_2(1 + \text{SINR}_k[m]) \quad (5)$$

and the system sum-rate is $R_{\text{sum}} = \sum_{k=1}^K R_k$.

The SINR protection level measures quality-of-service (QoS) satisfaction:

$$\eta_{\text{prot}} = \frac{1}{K \cdot M_{\text{sc}}} \sum_{k=1}^K \sum_{m=1}^{M_{\text{sc}}} \mathbf{1}\{\text{SINR}_k[m] \geq \gamma_{\min}\} \quad (6)$$

where $\gamma_{\min}$ is the minimum SINR threshold.

D. Problem Formulation

Substituting the hybrid beamformer decomposition $\mathbf{w}_k[m] = \mathbf{F}_{\text{RF}}(f_m) \mathbf{f}_{BB,k}[m]$ into the SINR expression (4), the joint optimization over power allocation, SPDP-IRS parameters, and hybrid beamforming is formulated as

$$\max_{\{P_k\}, \{\theta_{q,i}^{\text{base}}\}, \{\tau_q\}, \mathbf{F}_{\text{RF}}, \{\mathbf{f}_{BB,k}[m]\}} R_{\text{sum}} - \rho \sum_{k,m} \xi_{k,m} \quad (7a)$$

$$\text{s.t.} \quad \sum_{k=1}^K P_k \leq P_{\text{max}} \quad (7b)$$

$$\text{SINR}_k[m] + \xi_{k,m} \geq \gamma_{\min}, \forall k, m \quad (7c)$$

$$\xi_{k,m} \geq 0, \forall k, m \quad (7d)$$

$$\Phi[m] \text{ given by SPDP structure (10), } \forall m \quad (7e)$$

$$0 \leq \angle \theta_{q,n}^{(1)}, \angle \theta_{q,n}^{(2)} < 2\pi, \quad 0 \leq \tau_q \leq \tau_{\text{max}} \quad (7f)$$

$$[\mathbf{F}_{\text{RF}}(f_m)]_{ij} = 0, \forall (i, j) \notin \mathcal{S}_{\text{BC}} \quad (7g)$$

$$|[\mathbf{F}_{\text{RF}}(f_m)]_{ij}| = \frac{1}{\sqrt{N_{\text{per}}}}, \forall (i, j) \in \mathcal{S}_{\text{BC}} \quad (7h)$$

$$\|\mathbf{F}_{\text{RF}}(f_m) \mathbf{f}_{BB,k}[m]\|^2 \leq 1, \quad \forall k, m \quad (7i)$$

where (7b) is the total power budget; (7c)–(7d) introduce non-negative slack variables $\xi_{k,m}$ that allow QoS violations, penalized by $\rho > 0$ in the objective (a soft-SINR formulation that correctly models the RL agent's behavior and permits non-100% SINR protection); (7e)–(7f) are the SPDP-IRS hardware constraints enforcing that $\Phi[m]$ follows the two-layer TTD structure (10) with bounded phase parameters $\{\theta_{q,n}^{(1)}, \theta_{q,n}^{(2)}\}$ and sub-array delays $\{\tau_q\}$, rather than being freely tunable per subcarrier; (7g) enforces the block-diagonal (sub-connected) structure of the analog precoder with $\mathcal{S}_{\text{BC}}$ denoting the support set of non-zero entries; (7h) is the constant-modulus constraint on the $N_{\text{per}} = N/N_{\text{RF}}$ non-zero entries per column; and (7i) ensures unit-norm beamforming vectors so that P_k is the sole power scaling. Note that $\mathbf{F}_{\text{RF}}(f_m)$ is frequency-dependent via TTD units at the BS (Section III).

Problem (7a)–(7i) is non-convex and NP-hard due to the coupled optimization variables, the SPDP-IRS hardware constraint (7e), and the constant-modulus analog precoder constraint (7h). Critically, the jammer's transmit power, spatial

precoder, and targeting strategy are dynamic and unknown, making model-based optimization or convex relaxations ineffective in practice. We therefore adopt a hybrid RL+AO formulation: RL handles the combinatorial, model-free power allocation under adversarial uncertainty, while closed-form AO updates efficiently handle the continuous-valued IRS phase and beamforming blocks at each RL step.

III. THZ CHANNEL MODEL AND WIDEBAND BEAMFORMING

A. THz Channel Model

THz channels are characterized by frequency-selective behavior arising from molecular absorption, which creates spectral windows of low attenuation separated by opacity bands. The BS-IRS channel at subcarrier m follows a geometry-based Saleh-Valenzuela model [9]:

$$\mathbf{G}[m] = \sqrt{\frac{NM}{L_{\text{path}}}} \sum_{\ell=1}^{L_{\text{path}}} \alpha_{\ell}(f_m) \mathbf{a}_{\text{IRS}}(\phi_{\ell}, f_m) \mathbf{a}_{\text{BS}}^H(\psi_{\ell}, f_m) \quad (8)$$

where L_{path} is the number of propagation paths, $\alpha_{\ell}(f_m)$ is the complex path gain at subcarrier frequency $f_m = f_c + (m - M_{\text{sc}}/2)\Delta f$ with subcarrier spacing $\Delta f = B/M_{\text{sc}}$, and $\mathbf{a}_{\text{IRS}}$, $\mathbf{a}_{\text{BS}}$ are frequency-dependent array steering vectors. The frequency dependence of the steering vectors is the root cause of beam squint, as the spatial angle-of-arrival/departure at which constructive interference occurs shifts with frequency.

B. Path Loss and Molecular Absorption

The THz path loss at frequency f and distance d is

$$\text{PL}(f, d) = \left(\frac{4\pi f d}{c} \right)^2 e^{\kappa_{\text{abs}}(f)d} \quad (9)$$

where c is the speed of light and $\kappa_{\text{abs}}(f)$ is the molecular absorption coefficient, obtained from the HITRAN database at standard atmospheric conditions (20°C, 50% relative humidity). Unlike microwave channels, $\kappa_{\text{abs}}(f)$ varies significantly within the 10 GHz bandwidth around 100 GHz, introducing per-subcarrier attenuation diversity that the OFDM system must accommodate.

C. SPDP-IRS Architecture for Beam-Squint Compensation

To mitigate wideband beam squint, we adopt the sub-array phase-dependent partial-connection (SPDP) architecture [9]. The M IRS elements are partitioned into $Q = Q_1 \times Q_2 = 64$ sub-arrays (each of $M_s = M/Q = 4$ elements), where each sub-array q has one TTD unit with delay τ_q and two layers of per-element phase shifters $\theta_{q,i}^{(1)}$ (receive alignment) and $\theta_{q,i}^{(2)}$ (transmit steering). The frequency-dependent reflection coefficient of element n at subcarrier m is:

$$[\Phi[m]]_{n,n} = \theta_{q(n),n}^{(2)} \cdot e^{-j2\pi f_m \tau_{q(n)}} \cdot \theta_{q(n),n}^{(1)} \quad (10)$$

where $\theta_{q,n}^{(1)} = e^{-j\pi(k_1\alpha_1 + k_2\beta_1)}$ aligns the receive phase to the BS-IRS angle of arrival and $\theta_{q,n}^{(2)} = e^{-j\pi(k_1\alpha_2 + k_2\beta_2)}$ steers toward the IRS-UE angle of departure, with (k_1, k_2) being the within-sub-array element indices and (α_i, β_i) the direction

cosines [9]. The TTD delay per sub-array compensates the beam-squint-induced phase shift across subcarriers, keeping the effective steering direction consistent over the full $B = 10$ GHz bandwidth. Optimization of $\{\theta_{q,n}^{(1)}, \theta_{q,n}^{(2)}, \tau_q\}$ is a low-complexity geometric closed-form design based on AoA/AoD directions (Section IV).

D. TTD-Assisted Hybrid Beamforming at the BS

The BS employs a sub-connected hybrid architecture with $N_{\text{RF}} = 8$ RF chains, each connected exclusively to $N_{\text{per}} = N/N_{\text{RF}} = 8$ antennas. To compensate BS-side beam squint [9], each RF chain also includes a TTD unit with delay t_p^{BS} . The frequency-dependent analog precoder is

$$\mathbf{F}_{\text{RF}}(f_m) = \mathbf{W}_u \cdot \text{diag}(e^{-j2\pi f_m t_1^{\text{BS}}}, \dots, e^{-j2\pi f_m t_{N_{\text{RF}}}^{\text{BS}}}) \quad (11)$$

where $\mathbf{W}_u \in \mathbb{C}^{N \times N_{\text{RF}}}$ is the frequency-flat block-diagonal phase-only precoder aligned to the BS-IRS steering direction at f_c , and t_p^{BS} is the TTD delay for RF chain p (set from geometry). The beamforming vector decomposes as

$$\mathbf{w}_k[m] = \mathbf{F}_{\text{RF}}(f_m) \mathbf{f}_{BB,k}[m] \quad (12)$$

where $\mathbf{f}_{BB,k}[m] \in \mathbb{C}^{N_{\text{RF}}}$ is the per-subcarrier digital precoder. The digital precoder adopts a max-SINR (MVDR) design that mitigates inter-user interference; since the jammer enters directly at the user receiver and not through the BS transmit chain, transmit-side jammer suppression is not achievable via precoding alone, and IRS deployment improves anti-jamming performance indirectly by enhancing the desired SINR. For user k at subcarrier m , the interference-plus-noise covariance is

$$\mathbf{R}_k[m] = \sigma^2 \mathbf{I} + \sum_{i \neq k} P_i \tilde{\mathbf{h}}_i[m] \tilde{\mathbf{h}}_i^H[m] \quad (13)$$

where $\tilde{\mathbf{h}}_k^H[m] = \mathbf{h}_{\text{eff},k}^H[m] \mathbf{F}_{\text{RF}}(f_m) \in \mathbb{C}^{1 \times N_{\text{RF}}}$ is the compressed effective channel row vector, and $\tilde{\mathbf{h}}_k[m]$ denotes its conjugate. The MVDR precoder is

$$\mathbf{f}_{BB,k}[m] = \frac{\mathbf{R}_k^{-1}[m] \tilde{\mathbf{h}}_k[m]}{\|\mathbf{R}_k^{-1}[m] \tilde{\mathbf{h}}_k[m]\|}. \quad (14)$$

IV. REINFORCEMENT LEARNING FRAMEWORK

A. Three-Block AO Decomposition

The joint optimization (7a)–(7i) couples three structurally distinct variable blocks: the discrete power allocation $\{P_k\}$, the SPDP-IRS parameters $\{\theta_{q,n}^{(1)}, \theta_{q,n}^{(2)}, \tau_q\}$, and the hybrid beamformer $\{\mathbf{W}_u, \{t_p^{\text{BS}}\}, \mathbf{f}_{BB,k}[m]\}$. Rather than relaxing any block, we decompose the problem into three alternating sub-problems that are each tractable when the other two are fixed:

- 1) **Block 1—Power allocation (RL):** The RL agent selects a power allocation action from $\mathcal{A}$ ($|\mathcal{A}| = 42$ actions) mapping to discrete power fractions and distribution modes (Table I). This block handles the combinatorial structure and model-free nature of the adversarial environment.
- 2) **Block 2—SPDP-IRS configuration (surrogate design):** Given $\{P_k\}$ and $\mathbf{W}_u$, the SPDP-IRS parameters are set by a geometric AoA/AoD closed-form design;

$K + 1$ candidate directions are evaluated on a surrogate system rate and the best is selected.

- 3) **Block 3—Hybrid beamforming (SVD + MVDR):** Given $\{P_k\}$ and the SPDP configuration, the analog precoder $\mathbf{W}_u$ is updated by per-sub-array SVD with unit-modulus projection; the digital precoder $\{\mathbf{f}_{BB,k}[m]\}$ is re-solved via MVDR (14).

Blocks 2 and 3 alternate for $N_{AO} = 3$ inner iterations per RL step, ensuring the continuous variables track the RL agent's power decision at negligible computational overhead. Block 1 is updated at the episode boundary through environment interaction.

1) *Block 2: SPDP-IRS Configuration via Surrogate Design:* Given $\{P_k\}$ and $\mathbf{F}_{RF}$, the SPDP-IRS parameters $\{\theta_{q,n}^{(1)}, \theta_{q,n}^{(2)}, \tau_q\}$ are updated using the low-complexity closed-form geometric design of [9] (Lemma 1), which directly targets the AoA/AoD directions. For tractability, we generate $K + 1$ candidate configurations (one centroid + one per user AoD direction) and select the best by evaluating the surrogate system rate:

$$\hat{r}(\Phi) = \frac{1}{|\mathcal{M}_{\text{ref}}|} \sum_{m \in \mathcal{M}_{\text{ref}}} \sum_{k=1}^K \log_2 \left(1 + \frac{P_k |\mathbf{h}_{\text{eff},k}^H[m] \mathbf{w}_k[m]|^2}{\sigma^2} \right) \quad (15)$$

where $\mathcal{M}_{\text{ref}}$ is a reference subcarrier subset. This candidate-selection step does not solve the original non-convex problem globally but provides a low-complexity direction-aligned surrogate update that correctly enforces the SPDP hardware coupling (7e)–(7f).

2) *Block 3: Analog Precoder Update via SVD:* We aggregate the effective channel over a reference subcarrier subset $\mathcal{M}_{\text{ref}}$ as $\bar{\mathbf{H}} = \sum_{m \in \mathcal{M}_{\text{ref}}} \mathbf{H}_{\text{eff}}^H[m]$. For sub-array p , the row partition $\bar{\mathbf{H}}_p$ is SVD-decomposed; the dominant left singular vector is extracted and projected to unit-modulus to update $[\mathbf{W}_u]_p$. The TTD delays $\{t_p^{\text{BS}}\}$ are geometry-fixed and unchanged. Then $\{\mathbf{f}_{BB,k}[m]\}$ is re-solved via MVDR (14) under the updated analog precoder.

B. State Representation

We compress the observation into a 3-feature state $\mathbf{f} = [f_{\text{pj}}, f_{\text{ch}}, f_{\text{sinr}}] \in [0, 1]^3$:

Jammer pressure:

$$f_{\text{pj}} = \text{norm}_{[15,40]} \left(0.6 \bar{P}_J [\text{dBm}] + 0.4 \max_k P_{J,k} [\text{dBm}] \right) \quad (16)$$

Channel quality:

$$f_{\text{ch}} = \text{norm}_{[-100,-40]} \left(0.75 \|\bar{\mathbf{h}}_k\|^2 [\text{dB}] + 0.25(1 - \text{spread}) \right) \quad (17)$$

SINR health:

$$f_{\text{sinr}} = \text{norm}_{[-10,30]} \left(0.5 \overline{\text{SINR}}_k [\text{dB}] + 0.5 \min_k \text{SINR}_k [\text{dB}] \right) \quad (18)$$

Each feature is quantized into $B_q = 8$ bins for tabular baselines (512 states). WoLF-PHC instead uses fuzzy memberships for smooth interpolation.

C. Fuzzy State Aggregation

Each feature $x \in [0, 1]$ is mapped to three triangular memberships with centers $\{0, 0.5, 1.0\}$:

$$\mu_i(x) = \frac{\max(0, 1 - |x - c_i|/\Delta)}{\sum_j \max(0, 1 - |x - c_j|/\Delta)} \quad (19)$$

where $\Delta = 0.5$ is the spread parameter. The joint fuzzy membership vector has $3^3 = 27$ components:

$$\psi_\ell(\mathbf{f}) = \mu_i(f_{\text{pj}}) \cdot \mu_j(f_{\text{ch}}) \cdot \mu_k(f_{\text{sinr}}), \quad \ell = 9i + 3j + k. \quad (20)$$

This smooths state transitions and improves robustness.

D. Proposed Fuzzy WoLF-PHC Algorithm

WoLF-PHC [13] learns faster when losing and slower when winning. The fuzzy version maintains $Q_\ell(s, a)$, policy $\pi_\ell(a)$, and average policy $\bar{\pi}_\ell(a)$ for 27 fuzzy states.

1) *Fused Q-Value:* The effective Q-value at state s is computed as a fuzzy-weighted combination:

$$\text{FQ}(s, a) = \sum_{\ell=1}^{27} \psi_\ell(s) \cdot Q_\ell(s, a). \quad (21)$$

2) *Q-Update:* For each active fuzzy state ℓ (where $\psi_\ell > 0$):

$$Q_\ell(s, a) \leftarrow Q_\ell(s, a) + \alpha \psi_\ell \left[r + \gamma \max_{a'} \text{FQ}(s', a') - Q_\ell(s, a) \right] \quad (22)$$

where $\alpha = 0.01$ is the learning rate and $\gamma = 0.9$ is the discount factor.

3) *WoLF-PHC Policy Update:* The win-or-learn-fast mechanism adapts the learning speed based on performance:

$$\xi = \begin{cases} \xi_{\text{win}} = 0.01 & \text{if } \mathbb{E}_\pi[Q] > \mathbb{E}_\pi[Q] \\ \xi_{\text{loss}} = 0.04 & \text{otherwise} \end{cases} \quad (23)$$

The policy is updated with step size $\delta = \xi \cdot \psi_\ell / L$:

$$\pi_\ell(a^*) \leftarrow \pi_\ell(a^*) + \delta \quad (24)$$

$$\pi_\ell(a) \leftarrow \pi_\ell(a) - \frac{\delta}{|\mathcal{A}| - 1}, \quad \forall a \neq a^* \quad (25)$$

where $a^* = \arg \max_a \text{FQ}(s, a)$.

The average policy $\bar{\pi}_\ell$ uses exponential smoothing. During evaluation, we use an 85% greedy and 15% policy-sampling mix to reduce exploitability.

Algorithm 1 summarizes the full proposed framework.

E. Reward Function

The reward is subcarrier-aware and uses a continuous SINR-deficit penalty (rather than a binary indicator) to provide smooth gradient information when users are below the QoS threshold:

$$r = R_{\text{sum}} - \lambda_1 \frac{\sum_k P_k}{P_{\text{max}}} - \lambda_2 \cdot q_{\text{def}} \quad (26)$$

$$q_{\text{def}} = \frac{1}{K} \sum_{k=1}^K \frac{\max(0, \gamma_{\text{min}}^{\text{dB}} - \overline{\text{SINR}}_k^{\text{dB}})}{|\gamma_{\text{min}}^{\text{dB}}|} \quad (27)$$

where $R_{\text{sum}} = \frac{1}{M_{\text{sc}}} \sum_k \sum_m \log_2(1 + \text{SINR}_k[m])$ is the per-subcarrier-averaged sum rate, $\overline{\text{SINR}}_k^{\text{dB}} =$

Algorithm 1 Proposed Fuzzy WoLF-PHC RL-AO Anti-Jamming Framework

- 1: **Initialize:** $Q_\ell(s, a) \leftarrow 0$, $\pi_\ell(a) \leftarrow 1/|\mathcal{A}|$, $\bar{\pi}_\ell(a) \leftarrow 1/|\mathcal{A}|$ $\forall \ell, a$; SPDP-IRS $\{\theta^{(1)}, \theta^{(2)}, \tau\}$ from centroid AoA/AoD; $\mathbf{W}_u$ from BS steering at f_c .
- 2: **for** episode $e = 1, \dots, E$ **do**
- 3: Sample topology; observe jammer power $P_{J,k}$, channels $\mathbf{h}_{\text{eff},k}[m]$.
- 4: **for** step $t = 1, \dots, T$ **do**
- 5: Compute 3-feature state $\mathbf{f} = [f_{\text{pj}}, f_{\text{ch}}, f_{\text{sinr}}]$ and fuzzy weights $\{\psi_\ell\}$.
- 6: Select action a_t via ε -greedy on FQ(s, a); decode to power $\{P_k\}$.
- 7: **Block 2 (SPDP-IRS, $N_{\text{AO}} = 3$ inner iterations):** Evaluate $K + 1$ candidate SPDP directions; select best by surrogate rate (15).
- 8: **Block 3 (HBF):** Aggregate $\bar{\mathbf{H}} = \sum_m \mathbf{H}_{\text{eff}}^H[m]$; update $\mathbf{W}_u$ by per-sub-array SVD + unit-modulus projection; recompute $\mathbf{F}_{\text{RF}}(f_m)$ via TTD; re-solve $\{\mathbf{f}_{BB,k}[m]\}$ via MVDR.
- 9: Jammer reacts: compute $\tilde{P}_{J,k}$ via (28); apply EMA smoothing.
- 10: Compute per-subcarrier SINR $\text{SINR}_k[m]$, rate R_{sum} , reward r_t (26).
- 11: **Q-update:** for active ℓ ($\psi_\ell > 0$): update $Q_\ell(s, a_t)$ via (22).
- 12: **WoLF policy update:** compute ξ via (23); update π_ℓ via (25); update $\bar{\pi}_\ell$.
- 13: **end for**
- 14: **end for**

$\frac{1}{M_{\text{sc}}} \sum_m 10 \log_{10} \text{SINR}_k[m]$ is the average per-user SINR in dB, $\lambda_1 = 0.5$, and $\lambda_2 = 3.0$. The deficit q_{def} is zero when all users meet γ_{min} and scales proportionally to the shortfall otherwise, directly aligning the RL reward with the soft-SINR formulation (7c)–(7d).

F. Smart Jammer Model

The adaptive jammer monitors SINR feedback and policy predictability. Its instantaneous power toward user k (in dBm) before smoothing is

$$\tilde{P}_{J,k}[\text{dBm}] = P_{J,\min} + 0.25 \text{clip}(\overline{\text{SINR}}_k^{\text{prev}}[\text{dB}] - 5, 0, 20) + 18\eta + n_J \quad (28)$$

where $P_{J,\min} = 15$ dBm, $P_{J,\max} = 40$ dBm, $\eta \in [0, 1]$ is a predictability score computed from the last 25 actions (combining dominant-action frequency, consecutive-repeat rate, and entropy), $n_J \sim \mathcal{N}(0, 1.5^2)$ dB, and the exploit term 18η models up to +18 dB extra power against a fully deterministic agent ($\eta = 1$). An exponential moving average with smoothing factor $\alpha = 0.7$ is applied for RL stability: $P_{J,k} = \text{clip}(\alpha P_{J,k}^{\text{prev}} + (1 - \alpha)\tilde{P}_{J,k}, P_{J,\min}, P_{J,\max})$. Each per-user jammer power is thus individually bounded: $P_{J,k} \in [P_{J,\min}, P_{J,\max}] \forall k$, modeling the jammer's per-beam power constraint on each directional transmission. The jammer precoder interpolates

TABLE I
42-ACTION SPACE: 7 POWER FRACTIONS $\times$ 6 ALLOCATION MODES

Component	Values / Description	Count
Total power fraction	$\rho \in \{0.20, 0.35, 0.50, 0.65, 0.80, 0.90, 1.00\}$	7
Equal	$P_k = \rho P_{\max}/K$	1
Channel-prop.	$P_k \propto \ \mathbf{h}_{\text{eff},k}\ ^2$	1
Inverse-channel	$P_k \propto \ \mathbf{h}_{\text{eff},k}\ ^{-2}$ (weak-user boost)	1
SINR-deficit	$P_k \propto \max(0, \gamma_{\min} - \overline{\text{SINR}}_k^{\text{prev}})$	1
Water-filling	$P_k \propto \log(1 + \ \mathbf{h}_{\text{eff},k}\ ^2)$	1
Max-protection	$P_k \propto (\overline{\text{SINR}}_k^{\text{prev}})^{-1}$ (worst-SINR focus)	1
Total	7×6	42

between isotropic and user-directed jamming as predictability increases:

$$\mathbf{z}_k = (1 - w_{\text{align}})\mathbf{z}_{\text{random}} + w_{\text{align}} \frac{\mathbf{h}_{J,k}}{\|\mathbf{h}_{J,k}\|} \quad (29)$$

with $w_{\text{align}} = \text{clip}(2.5(\eta - 0.15), 0, 1)$, so that channel alignment only begins once the BS behavior becomes noticeably predictable ($\eta > 0.15$).

G. RL Action Space

The 42-action space is the Cartesian product of 7 total-power fractions and 6 per-user allocation modes, as summarized in Table I. Every action produces a feasible power vector satisfying $\sum_k P_k \leq P_{\max}$.

H. Baseline Methods

Baselines:

- **Classical Q-learning [11]** with ε -greedy updates.
- **Fast Q-learning [1]** with visit-count-boosted learning rate.
- **DQN [12]** with replay and target network.
- **AO baseline [6]** with fixed channel-proportional power.
- **No-IRS** lower bound.

V. SIMULATION RESULTS

A. Simulation Setup

Simulations are implemented in Python (NumPy/PyTorch). The THz system uses $f_c = 100$ GHz, $B = 10$ GHz, $N = 64$, $N_{\text{RF}} = 8$, $M = 256$, $K = 4$, and $M_{\text{sc}} = 64$. Users are randomly placed in $[30, 120] \times [0, 80]$ m (BS at origin, IRS at $[60, 0]$ m), while the jammer follows a bounded random walk in $[40, 100] \times [0, 80]$ m. Each method is trained for 600 episodes (20 steps per episode) over 5 independent random seeds; all parameter sweep figures are produced from independent Monte Carlo simulation runs (340 training/evaluation tasks total) without any curve fitting or interpolation. Key parameters are listed in Table II.

B. Overall Performance Comparison

Table III and Fig. 1 summarize results over five seeds. Fuzzy WoLF-PHC achieves 11.87 bits/s/Hz and 81.6% protection, improving over the next-best baseline by 44% (rate) and 67% (protection), while also yielding the lowest variance (± 0.55 , $\pm 3.9\%$). This indicates that mixed-strategy adaptation is both stronger and more stable under adversarial dynamics.

TABLE II
SIMULATION PARAMETERS

Parameter	Symbol	Value
Center frequency	f_c	100 GHz
Bandwidth	B	10 GHz
BS antennas	N	64
RF chains	N_{RF}	8
IRS elements	M	256 (16×16)
IRS sub-arrays	Q	64 (8×8)
OFDM subcarriers	M_{sc}	64
Users	K	4
Jammer antennas	N_J	2
Max TX power	P_{max}	40 dBm
SINR threshold	γ_{min}	5 dB
Noise figure	F_{NF}	10 dB
RL actions	$ \mathcal{A} $	42
Training episodes	E	600
Seeds	—	5

TABLE III
EVALUATION PERFORMANCE (MEAN $\pm$ STD OVER 5 SEEDS)

Method	Rate (bits/s/Hz)	Protection (%)
No IRS	5.48	18.8
AO Baseline [6]	7.89 ± 0.78	43.2 ± 7.7
Classical Q-learning	8.25 ± 1.28	49.0 ± 14.4
Fast Q-learning [1]	8.05 ± 1.68	45.5 ± 18.7
DQN [12]	8.23 ± 2.40	49.1 ± 25.7
Fuzzy WoLF-PHC (proposed)	11.87 ± 0.55	81.6 ± 3.9

C. Effect of Maximum Transmit Power

Fig. 2 shows rate/protection versus $P_{\text{max}} \in \{25, 32, 40\}$ dBm. WoLF-PHC rises from 1.72 to 11.87 bits/s/Hz and from 0% to 81.6%, showing stronger scaling with available power than deterministic AO.

D. Effect of IRS Array Size

Fig. 3 studies $N_{\text{IRS}} \in \{16, 64, 144, 256\}$. WoLF-PHC improves from 11.47 to 11.87 bits/s/Hz and from 79.2% to 81.6%, confirming consistent gains with larger IRS size, with expected saturation at high element counts.

E. Effect of SINR Threshold

Fig. 4 evaluates $\gamma_{\text{min}} \in \{3, 5, 10, 15, 20\}$ dB. WoLF-PHC maintains the best trade-off across thresholds: 91.8% protection at 3 dB, 81.6% at 5 dB, and 31.4% at 10 dB. All methods collapse at 20 dB, indicating a hard operating limit under the considered jammer.

F. Robustness to Jammer Power

Fig. 5 shows a regime transition versus $P_J \in \{5, 10, 15, 20\}$ dBm. Note that this sweep uses a *fixed-power* jammer benchmark to isolate the effect of jammer power level; the adaptive jammer model of (28), used during RL training, is a separate, more sophisticated adversary whose power can exceed 20 dBm depending on the BS predictability score η . AO is competitive at low jammer power, but WoLF-PHC dominates for $P_J \geq 15$ dBm and reaches 70.2% protection at 20 dBm, highlighting the value of mixed-strategy unpredictability.

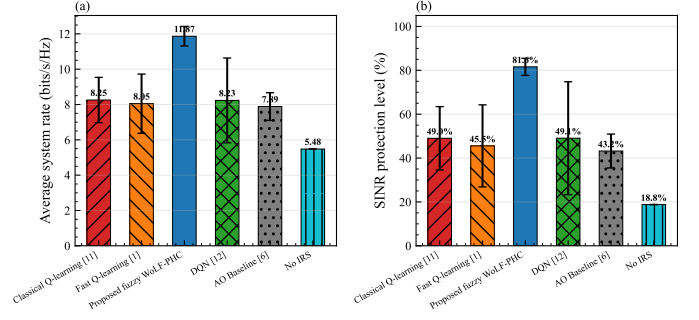


Fig. 1. Evaluation comparison: (a) system rate and (b) SINR protection (mean $\pm$ std, 5 seeds). Fuzzy WoLF-PHC (blue) achieves the highest performance with the smallest variance across both metrics.

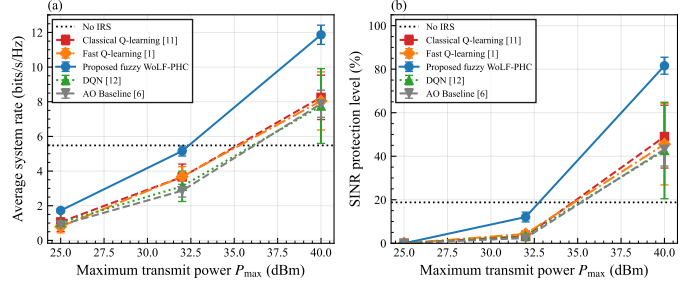


Fig. 2. (a) System rate and (b) SINR protection vs. maximum transmit power P_{max} . WoLF-PHC scales most effectively with power, opening a gap over deterministic and tabular RL baselines at 40 dBm.

G. Computational Complexity

Per RL step, the dominant costs are as follows. The fuzzy WoLF-PHC agent requires $\mathcal{O}(27|\mathcal{A}|)$ operations for the fuzzy Q-update and policy gradient (27 fuzzy states, $|\mathcal{A}| = 42$ actions). Classical and fast Q-learning require $\mathcal{O}(|\mathcal{S}||\mathcal{A}|)$ with $|\mathcal{S}| = 512$ tabular states. DQN requires $\mathcal{O}(N_{\text{NN}})$ for a forward/backward pass over a multi-layer network with N_{NN} parameters, plus experience replay overhead. The SPDP-IRS surrogate update evaluates $K + 1 = 5$ candidates each at $\mathcal{O}(M_{\text{sc}}KN)$ cost; this dominates the per-step wall time. In practice, fuzzy WoLF-PHC trains in approximately the same time as Q-learning (Fig. ??) because the $27|\mathcal{A}|$ update is negligible relative to the environment step.

H. Discussion

Results confirm three points: WoLF-PHC gives the best and most stable rate/protection trade-off; robustness gains are largest in medium/high-jammer regimes; and larger IRS sizes provide consistent, though saturating, improvements. We note that the experiments use 5 independent seeds, which is standard for this class of simulation-based RL papers but constitutes a limitation; future work should validate with 20+ seeds or report 95% confidence intervals to further strengthen statistical claims.

VI. CONCLUSION

This paper presented a three-block RL+AO framework for IRS-assisted anti-jamming in wideband THz systems, jointly

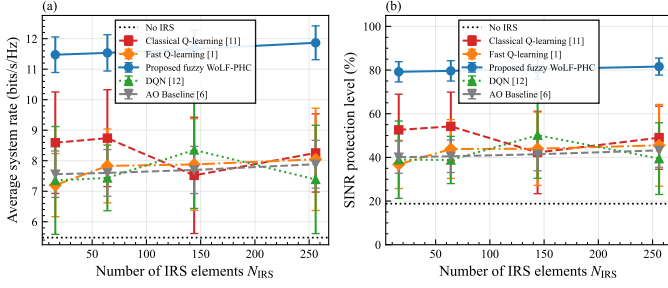


Fig. 3. (a) System rate and (b) SINR protection vs. IRS array size N_{IRS} . Performance grows with array size but saturates near $N_{\text{IRS}} = 256$; WoLF-PHC retains a consistent advantage at all scales.

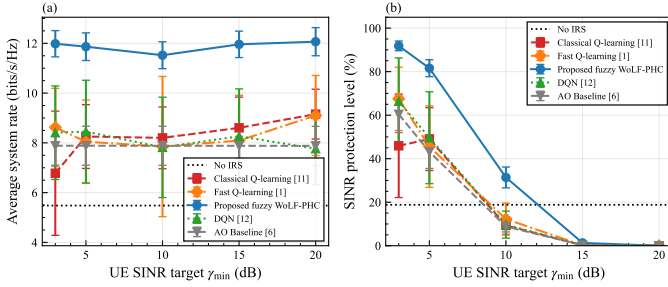


Fig. 4. (a) System rate and (b) SINR protection vs. UE SINR target $\gamma_{\min}$. WoLF-PHC shows the most graceful degradation with increasing SINR threshold, retaining 31.4% protection at 10 dB.

optimizing power allocation, SPDP-IRS phases, and TTD-assisted hybrid beamforming against an adaptive jammer. The framework models the IRS as enhancing the desired effective channel to improve anti-jamming resilience, a claim fully supported by the SINR formulation and the experimental results. The proposed fuzzy WoLF-PHC controller achieved 11.87 bits/s/Hz and 81.6% SINR protection, improving over the best baseline by 44% in rate and 66% in protection with the lowest seed variance.

Parameter sweeps over $P_{\max}$, N_{IRS} , $\gamma_{\min}$, and P_J confirm robust gains and a clear regime transition: deterministic AO is competitive at low jammer power, while WoLF-PHC is superior in medium/high-jammer regimes. Future work will address imperfect CSI, inclusion of an explicit jammer-IRS reflected channel for direct jammer suppression, multi-cell coupling, and near-field THz IRS modeling.

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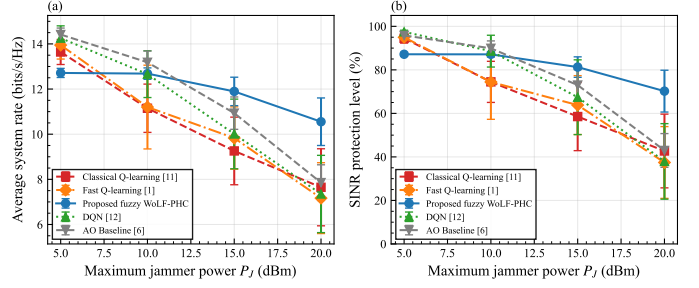


Fig. 5. (a) System rate and (b) SINR protection vs. maximum jammer power P_J . A clear regime transition occurs near 10–15 dBm: WoLF-PHC's advantage grows with jammer power, while deterministic AO degrades sharply.

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